
Prefrontal Meta-Control Enables the Improved Adaptivity of Reinforcement Learning Agents

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Abstract

Recent advances in reinforcement learning (RL) have led to remarkable achievements in developing agents with super-human performance. While these agents predominantly rely on model-free (MF) principles, there is mounting interest in model-based (MB) approaches for their potential benefits in sample efficiency and adaptation speed. However, recent studies indicate that MB methods do not consistently outperform MF approaches and often struggle with tasks that humans find relatively straightforward. Decision neuroscience offers compelling insights into resolving these limitations. The brain employs a meta-control system, centred in the prefrontal cortex. This arbitration mechanism enables dynamic transitions between MB control when deliberate planning is warranted and MF control when habitual responses are more computationally efficient. Rooted from these neuroscientific principles, we propose Meta-Dyna - a novel RL agent that mirrors the brain's meta-control mechanisms. Meta-Dyna addresses key limitations of current RL systems by incorporating neural mechanisms for flexible, adaptive behaviour. The result of simulations on two different tasks, (i) *2-stage Markov decision task* well known in decision neuroscience and (ii) *GridworldLoCA* known as a benchmark environment for MB RL algorithms, confirms that our proposal is superior with respect to reward, choice optimality and training time to those of baseline RL algorithms. By incorporating neurally-inspired meta-control mechanisms, Meta-Dyna represents a significant step towards more robust and adaptable RL agents that can match human-level flexibility in dynamic environments

Keywords: Meta-Dyna; Meta Control; Arbitration; Model-Based and Model-Free; Reinforcement Learning

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1 Introduction

Deep Reinforcement Learning (Deep RL), which integrates RL with deep learning architectures, has demonstrated exceptional capabilities that surpass human performance across numerous domains. Whilst the predominant implementation of RL agents has relied upon model-free (MF) principles, there exists an expanding collection of model-based (MB) algorithms that aim to leverage enhanced sample efficiency and adaptive capacity.

Nevertheless, recent benchmark analyses challenge the assumption of MB supremacy over MF approaches. Contemporary research in decision neuroscience presents compelling evidence that suggests the fundamental principle underlying human RL resides in meta-control mechanisms – specifically, the arbitration between MB and MF learning strategies. This mechanism enables humans to adapt reliably to frequent environmental changes whilst optimising the balance between performance, efficiency and processing speed.

Rooted upon these neuroscientific insights, we propose a novel meta-control RL agent (Figure 1-A) that demonstrates rapid adaptation to environmental dynamics whilst managing variable goal states and state-transition uncertainties. This architectural framework implements prefrontal meta-control mechanisms, which in turn optimises task performance with limited experiences.

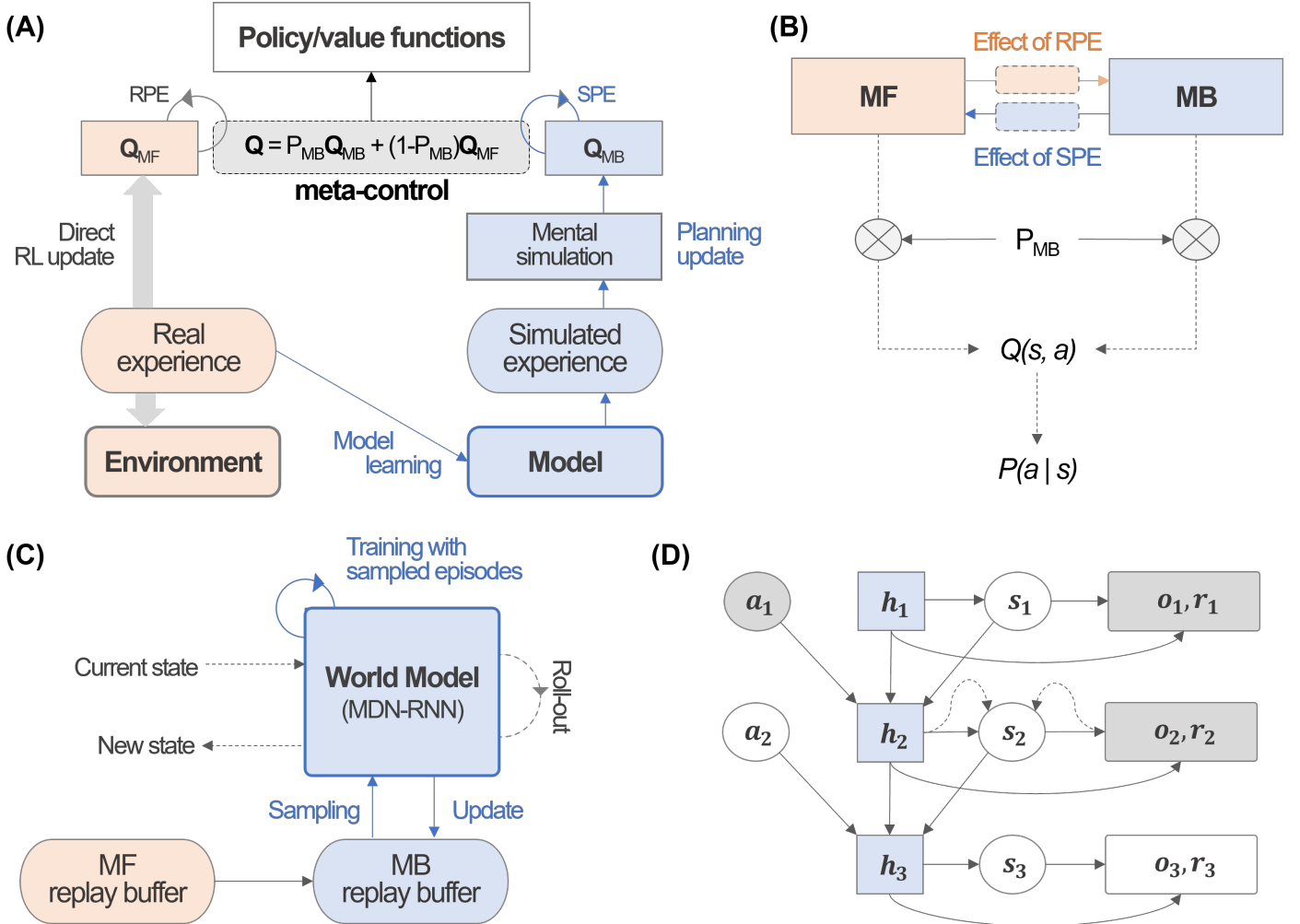


Figure 1: *Meta-Dyna* Overview. (A) Schematic architecture of *Meta-Dyna*. (B) An illustration of prefrontal meta-control, which arbitrates between MB and MF learning strategies. (C) A schematic representation of the world model that enables mental simulation. (D) A demonstration of the roll-out process, which implements mental simulation for given state-action pairs.

2 Meta-control RL Framework

According to Lee and colleagues (Lee, Shimojo, & O’Doherty, 2014), uncertainty in prediction performance for each learning strategy, quantified through prediction errors (PEs), serves as the crucial variable for implementing arbitration control. This mechanism manifests in specific scenarios: when the MB RL system exhibits low state-prediction errors (SPEs) whilst the MF RL system si-

multaneously shows high reward-prediction errors (RPEs), the MB RL demonstrates superior reliability. Consequently, behavioural choices become predominantly influenced by MB RL, whilst the impact of the MF RL system diminishes.

Should the environment shift rapidly towards heightened instability, whereby the MB RL’s predictive reliability falls below that of MF RL, behavioural influence transitions towards MF RL dominance. This adaptive process relies on the brain’s computation of strategy reliability through PEs, which generates a model choice probability P_{MB} that indicates strategic preference. The final decision-making process integrates Q-values from both MB and MF strategies, weighted by P_{MB} , to determine optimal policy selection within the environmental context (Lee et al., 2014; Kim, Park, O’Doherty, & Lee, 2019).

We implemented the mechanism of prefrontal meta-control using Dyna-Q (Sutton, 1990). So we named it Meta-Dyna (i.e., meta-control RL). As widely known, Dyna-Q combines Q-learning with Planning. it updates the Q-value in the MB manner by planning with the world model learned from experiences collected by direct interaction with the environment.

Extending from Dyna-Q, meta-control RL equipped two separate Q-values for both planning and Q-learning which represents the Q-values for MB (Q_{MB}) and MF (Q_{MF}), respectively. Using these, the arbitrator of meta-control RL computed P_{MB} using the reliability of each learning system based on PEs while training. The meta-control RL brings about the integrated Q-value using P_{MB} , by computing the weighted sum of each system (P_{MB} for Q_{MB} and $1 - P_{MB}$ for Q_{MF} ; Figure 1B).

3 Experiments

To precisely examine the performance, we conducted simulations on two different tasks where both goals and environmental structures (e.g., state-transition probability) change frequently. The first task is the 2-stage Markov Decision Task (MDT), which is designed to investigate the characteristics of human reinforcement learning in decision neuroscience (Lee et al., 2014; Gläscher, Daw, Dayan, & O’Doherty, 2010). The second task is GridworldLoCA, a widely known benchmark environment for model-based reinforcement learning (MB RL) algorithms in machine learning (Wan et al., 2022).

We hypothesize that meta-control RL would successfully adapt to frequent environmental changes relatively quicker than baseline RL algorithms. In our experiments, we employed Dyna-Q and FORWARD (a baseline of our proposal) for MB RL, and Q-Learning for model-free reinforcement learning (MF RL).

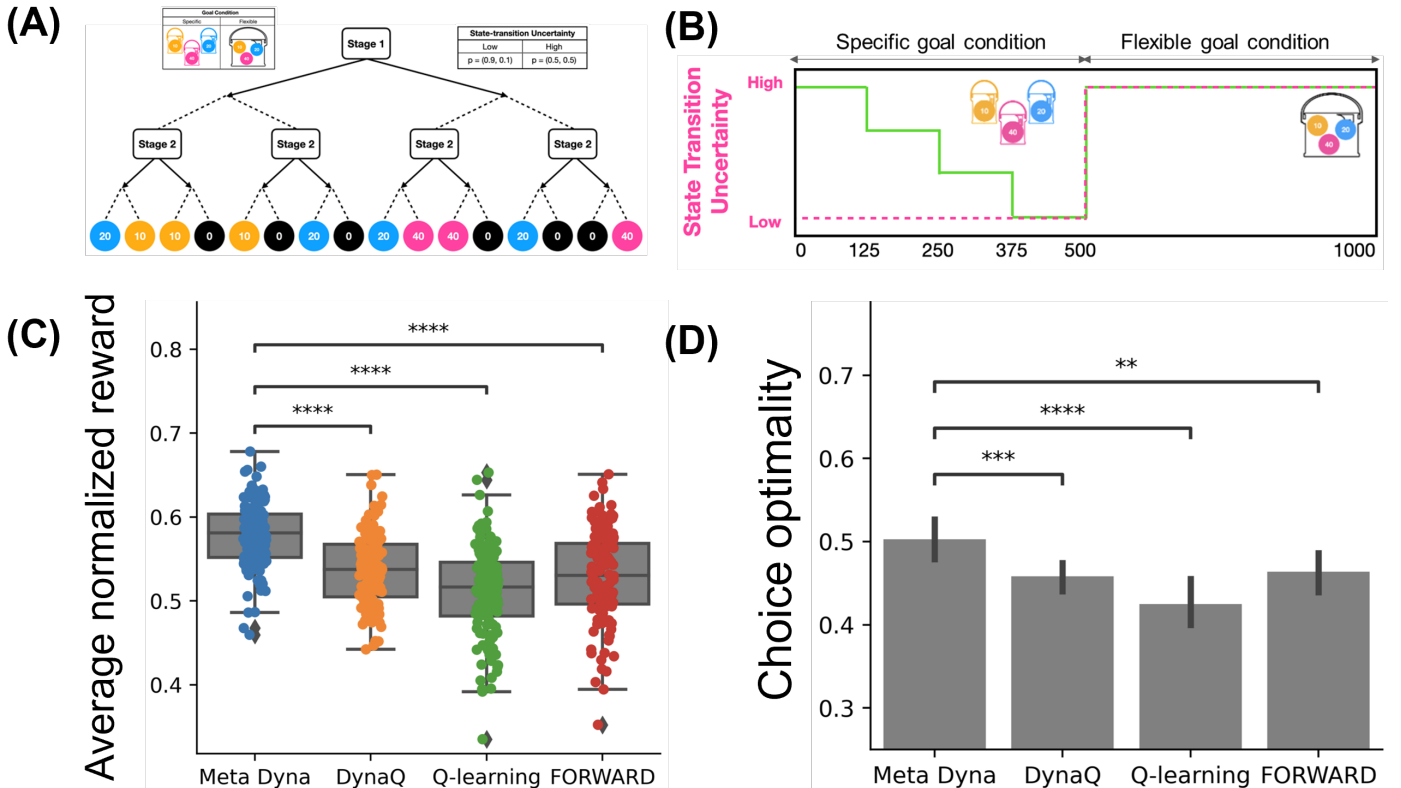


Figure 2: Experiment 1 - two stage Markov decision task with meta-control RL framework. (a) 2-stage MDT. (b) A simulation setting. (c) Average normalised reward. (d) Choice optimality (sample paired t -test; **: $p < 1e-2$, ***: $p < 1e-3$, ****: $p < 1e-4$)

In Experiment 1, we used the 2-stage Markov Decision Task (MDT) (Figure 1A) and simultaneously manipulated two task variables: state-transition uncertainty and goal condition. We used two sets of transition probabilities for the choice action: (0.5 vs 0.5) for the flexible goal condition and (0.9 vs 0.1) for the specific goal condition.

Each simulation session consisted of seven blocks. Within each block, the goal was changed every 125 steps and the uncertainty was changed every 500 steps (Figure 1B). We performed 100 sessions in total.

The results showed that the proposed meta-control reinforcement learning (RL) algorithm performed statistically better than the other RL algorithms. Meta-control RL achieved significantly higher average normalized rewards (Meta-Dyna: 0.61, Dyna-Q: 0.55, Q-learning: 0.52, FORWARD: 0.54; $p < 0.0001$, independent t-test) and choice optimality compared to the baselines, especially during frequent environmental changes. This finding directly implies that meta-control RL has a better adaptation capacity than the baselines.

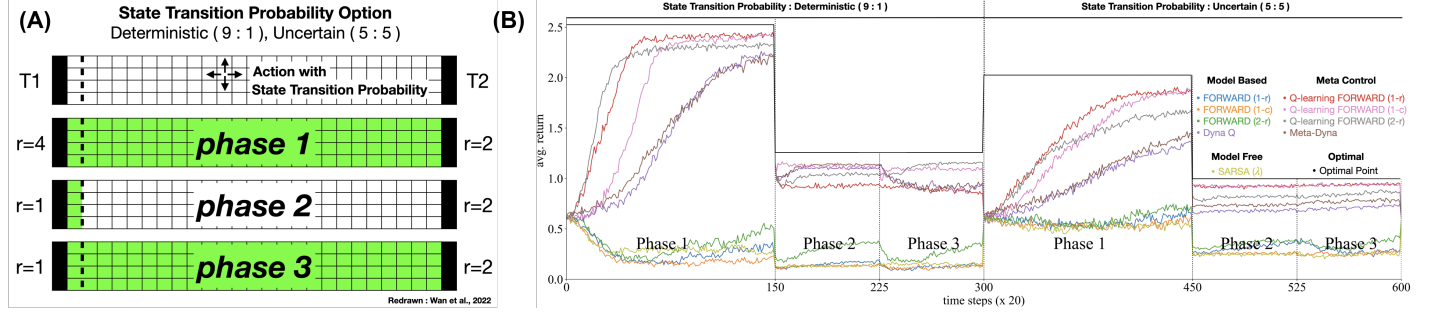


Figure 3: Experiment 2 (a) *GridworldLoCA* with changes in state transition probability. (b) Average return in the change of environmental settings.

In Experiment 2, we used the *GridworldLoCA* task to benchmark the performance of our proposal with that of model-based reinforcement learning (MB RL) algorithms in terms of sample efficiency under environmental changes. We manipulated two task variables: the reward ($r = 4, 2, 1$) and the initial state of the agent (Green) (Figure 2A).

Each simulation session consisted of two phases. The agent was required to perform the task from phase 1 to phase 2 in sequence, and this process was repeated twice per session. The initial states varied between phases, with the green area indicating the allowed starting positions for the agent in each phase. The reward setting also changed at each phase.

To increase the task’s difficulty and complexity, we introduced an additional task variable: the state-transition probability. Similar to Experiment 1, we used two sets of state-transition probabilities: (0.9, 0.1) and (0.5, 0.5). We performed ten simulation sessions in total.

We evaluated three types of agents in this experiment: a standalone model-based reinforcement learning (MB RL) agent (FORWARD), a standalone model-free reinforcement learning (MF RL) agent (Q-learning), and meta-control models (a combination of FORWARD and SARSA, and the meta-control Dyna-Q used in Experiment 1). We considered FORWARD as an optimal model and used its performance as an upper bound.

The results (Figure 2B) showed that the standalone agents failed to achieve the goal and adapt to the environment within the given episode horizon, exhibiting poor performance. In contrast, the meta-control agents quickly adapted to the environmental changes (from phase 1 to phase 3 repeatedly) within the same episode horizon.

Our proposed model outperformed Dyna-Q in terms of average return ($p < 2e - 4$, sample paired t-test) and time to success. Overall, incorporating meta-control in RL significantly improves adaptability compared to baseline RL algorithms.

4 Conclusion

Based upon inspiration from recent neuroscientific discoveries, we have developed meta-control RL framework, a novel RL architecture that incorporates prefrontal meta-control mechanisms. This innovative approach empowers RL agents with the capacity to rapidly adapt to dynamic environments whilst requiring comparatively minimal experience.

To rigorously evaluate the efficacy of meta-control RL, we conducted a series of comprehensive experimental simulations across two distinct paradigms: the two-stage Markov Decision Task (MDT), which is widely employed in decision neuroscience research and *GridWorldLoCA*, an established benchmark suite for model-based RL algorithms.

The experimental results unequivocally demonstrate that meta-control RL exhibits superior performance in comparison to baseline RL algorithms across multiple key metrics, including average reward acquisition, choice optimality, and the number of trials required for success. By virtue of its meta-control mechanisms, which dynamically modulate the integration of model-based and model-free

learning strategies, meta-control RL achieves expeditious adaptation to environmental perturbations, even under the constraints of limited training episodes.

Through this multifaceted research endeavour, we aspire to make substantive contributions to both the advances in computational RL and the elucidation of the neural underpinnings of human learning and decision-making. The insights gleaned from the development and analysis of meta-control RL hold the potential to inform the design of RL agents that more closely mirror human cognitive processes, thereby paving the way for the creation of autonomous systems that exhibit greater adaptability, efficiency, and robustness in the face of real-world challenges.

References

- Chalnick, A., & Billman, D. (1988). Unsupervised learning of correlational structure. In *Proceedings of the tenth annual conference of the cognitive science society* (pp. 510–516). Hillsdale, NJ: Lawrence Erlbaum Associates.
- Feigenbaum, E. A. (1963). The simulation of verbal learning behavior. In E. A. Feigenbaum & J. Feldman (Eds.), *Computers and thought*. New York: McGraw-Hill.
- Gläscher, J., Daw, N., Dayan, P., & O’Doherty, J. P. (2010). States versus rewards: dissociable neural prediction error signals underlying model-based and model-free reinforcement learning. *Neuron*, 66(4), 585–595.
- Hill, J. A. C. (1983). A computational model of language acquisition in the two-year old. *Cognition and Brain Theory*, 6, 287–317.
- Kim, D., Park, G. Y., O’Doherty, J. P., & Lee, S. W. (2019). Task complexity interacts with state-space uncertainty in the arbitration between model-based and model-free learning. *Nature communications*, 10(1), 5738.
- Lee, S. W., Shimojo, S., & O’Doherty, J. P. (2014). Neural computations underlying arbitration between model-based and model-free learning. *Neuron*, 81(3), 687–699.
- Matlock, T. (2001). *How real is fictive motion?* (Doctoral dissertation). Psychology Department, University of California, Santa Cruz.
- Newell, A., & Simon, H. A. (1972). *Human problem solving*. Englewood Cliffs, NJ: Prentice-Hall.
- Ohlsson, S., & Langley, P. (1985). *Identifying solution paths in cognitive diagnosis* (Tech. Rep. No. CMU-RI-TR-85-2). Pittsburgh, PA: Carnegie Mellon University, The Robotics Institute.
- Shrager, J., & Langley, P. (Eds.). (1990). *Computational models of scientific discovery and theory formation*. San Mateo, CA: Morgan Kaufmann.
- Sutton, R. S. (1990). Integrated architectures for learning, planning, and reacting based on approximating dynamic programming. In *Machine learning proceedings 1990* (pp. 216–224). Elsevier.
- Wan, Y., Rahimi-Kalahroudi, A., Rajendran, J., Momennejad, I., Chandar, S., & Van Seijen, H. H. (2022). Towards evaluating adaptivity of model-based reinforcement learning methods. In *International conference on machine learning* (pp. 22536–22561).